expect for a nonrelativistic particle. In other words, it's not $\mathbf{P}^2/2m$. This simpler example is worth exploring first, just to see how the mathematical apparatus works.

How do we represent this example in terms of wave functions $\psi(x)$ in the position basis? We'll start by plugging our operators into the time-dependent Schrödinger equation (Eq. 9.1):

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = -ci\hbar\frac{\partial\psi(x,t)}{\partial x}.$$

Notice that we're now writing ψ as a function of both *x and t*. Canceling the $i\hbar$ terms gives us

$$\frac{\partial\psi(x,t)}{\partial t} = -c\frac{\partial\psi(x,t)}{\partial x}, \tag{9.3}$$

which is a pretty simple equation. In fact, any function of $(x - ct)$ is a solution. By "function of $(x - ct)$," I mean any function that depends not on x and t separately, but only on the combination $(x-ct)$. To see how this works, just consider an arbitrary function $\psi(x - ct)$ and look at its derivatives. If you take the partial with respect to x, you just get

$$\frac{\partial\psi(x-ct)}{x}$$

because the derivative of $(x - ct)$ with respect to x is 1. But if you take the partial with respect to t, you get

$$-c\frac{\partial\psi(x-ct)}{\partial t}.$$